

ABISHEK S

+91 93427 64046 abisheksridharan.work@gmail.com  Portfolio  GitHub  LinkedIn  Kaggle

PROFESSIONAL SUMMARY

AI Engineer with 1.5 years of experience designing and shipping production LLM systems and data-intensive backends. At Pothys Retail, builds and runs the company's finance operations platform, spanning reconciliation, GST compliance, enterprise email and AI agents. Previously at CobuildX.ai, delivered four AI-powered SaaS products, including a Claude-integrated MCP server, a legal RAG platform and an open-source package on PyPI. Works across Python, TypeScript/React, AWS and GCP.

WORK EXPERIENCE

Pothys Retail Pvt Ltd

Jul 2026 – Present

AI Engineer

PRPL Finance Operations Platform

FastAPI | React 19 | TypeScript | PostgreSQL | GCP

Built and maintain the company's finance automation platform, **used daily by the finance team** across **~20 reconciliation and GST modules** (AP, AR, GST, TDS, GR/IR, inter-company, bank, card, tender and gift card). Own the architecture, backend and frontend development, role-based access control, security hardening, **500+ automated tests** and production deployment on Google Cloud.

GST Input Tax Credit Reconciliation

Python | Pandas | SAP OData | ReportLab

Designed an 11-level matching engine that reconciles **~3.9 lakh SAP purchase records against GSTR-2B in minutes**, covering exact, date, tax, invoice-similarity and credit-note rules. Integrated the live SAP vendor master over OData to generate a GSTR-1 non-filing notice for each vendor and **email ~2,000 vendors automatically**, replacing manual follow-up and protecting the company's input tax credit.

Enterprise Email Integration

Microsoft Graph API | OAuth 2.0 | PostgreSQL

Migrated every outgoing email, including weekly statements to **~92 accountants**, vendor notices, user invitations and system alerts, to a **Microsoft 365 shared mailbox via the Microsoft Graph API** with certificate-based authentication. Implemented rate limiting that **keeps sending within the organisation's hourly and daily limits**, with automatic pause and resume for large batches and support for large attachments.

LLM Finance Assistant and Agents

LLM Function Calling | Agentic AI | FastAPI

Built a conversational **LLM assistant** that answers finance questions from **live reconciliation data through function calling** and prepares actions such as sending reports, which **run only after the user confirms them**. Designed scheduled agents, including a **maker/checker agent** in which one model drafts findings and a second independently verifies them against the source data.

AP Accountant Performance Analytics

Python | Pandas | PostgreSQL

Automated weekly tracking of **~58,000 open vendor items** to show what each accountant cleared, carried forward or newly received, **matching the finance team's manual result exactly**. Scores and ranks **~98 accountants** on ageing reduction, offset clearance and task completion, with an AI-written management summary and a personalised pending-work email for each accountant.

Bank Statement Processing Engine

Python | Pandas | LLM Column Mapping

Replaced five bank-specific parsers with one configuration-driven engine for PDF, Excel and CSV statements, including tables without ruled lines, with totals cross-checked against each statement's summary. Added **AI-proposed column mapping** for new formats from masked sample rows, validated on the full file and **approved by a user before use** (identical totals on 54 real PhonePe files).

CobuildX.ai

Apr 2025 – Jun 2026

AI Engineer

Lead Management MCP Server

AWS Lambda | MCP | Claude | EventBridge

Built a **Model Context Protocol server on AWS Lambda** exposing **8 tools to Claude** for B2B sales automation: lead management, CRM filtering, ideal-customer-profile scoring (0 to 100) and contact enrichment through Apify with a Hunter.io fallback. Added a scheduled EventBridge worker for Slack reminders that **removed the manual steps from sales outreach**.

truAI: Legal RAG Platform

FastAPI | React | LangChain | AWS Lambda

Delivered a **DPDPA-compliant legal research platform** that streams answers with **sub-second first-token latency**, grounded in citations that separate binding statutes from non-binding guidance. Included an automated GitHub scanner for 300+ PII and cookie-compliance patterns, **reducing manual legal review effort by ~40%**.

truScanner (open source, PyPI)

Python | AWS Bedrock | Ollama

Built and published an open-source CLI on PyPI (v0.2.10) that scans codebases for 300+ patterns of personal and financial data, such as social security numbers, card numbers and API keys, with a three-tier LLM fallback (AWS Bedrock, Ollama, local quantised models) so it runs fully offline.

WarpX: AI Collaborative Workspace

React | FastAPI | Supabase

Designed a two-level LLM memory architecture that summarises each thread into a channel-wide context store, so the assistant keeps long-term context while cutting per-message token cost by ~60% through summary caching.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Backend & APIs	FastAPI, REST APIs, MCP (JSON-RPC 2.0), Microsoft Graph API, SAP OData, OAuth 2.0, JWT, Pydantic, SQLAlchemy, Alembic
AI & LLMs	LLM Agents & Function Calling, RAG, LangChain, Anthropic Claude API, AWS Bedrock, Google Vertex AI, Vector Embeddings, Ollama, PyTorch, TensorFlow
Frontend	React 18/19, TypeScript, Vite, Tailwind CSS, React Query
Databases	PostgreSQL, DynamoDB, Supabase, ChromaDB
Cloud & DevOps	GCP Compute Engine, AWS Lambda, API Gateway, EventBridge, nginx, Docker, Linux, GitHub Actions
Testing & Tools	Pytest, Locust (load testing), Git, ReportLab, PyPI publishing

EDUCATION

Mepco Schlenk Engineering College, B.Tech in Artificial Intelligence and Data Science

Oct 2021 – Apr 2025

PUBLICATIONS & OPEN SOURCE

- "YOLO-ASCA: A Rule-Based Framework for Identifying Safety Risks in Construction Management", WiSPNET 2025, SSN College of Engineering, Chennai (IEEE Xplore).
- truScanner v0.2.10 on PyPI, an open-source CLI for PII and financial data detection: pypi.org/project/truscanner

Kaggle Datasets (kaggle.com/abishek9324, 4 datasets, 500+ downloads)

Agent Failure Atlas 2026

Synthetic Benchmark | Agentic AI

Reproducible benchmark of multi-step AI-agent trajectories, published as a 9-table relational dataset with 2,202 runs and 25,964 steps and a 30-category failure taxonomy that records each failure, diagnosis and recovery attempt; the most downloaded of the four.

Fuzzy Invoice Reconciliation

Finance | Matching Algorithms

Accounts-payable dataset for invoices whose references and amounts do not line up exactly, with a solver that applies exact, fuzzy and subset-sum matching in sequence and recovers 94.4% of true matching pairs.

Vendor Subset-Sum Matching

Finance | Algorithms

Dataset for matching single vendor payments to combinations of open invoices, with a meet-in-the-middle solver that cuts search complexity from $O(2^n)$ to $O(2^{(n/2)})$ and reaches 99.2% precision and 97.5% recall.

Student Performance & Placement Prediction

Machine Learning | Education

Dataset of 1,500 students with causally related features (attendance, study habits, coding practice, internships) and two targets, final CGPA and campus placement, covering regression, classification and clustering in one set.